

WRITE-UP ON RESEARCH PROJECT FOR ANVESHAN 2024

Title of the Research Project: Design of IoT Architecture for Battery Charging Management Technology

Team Members:

S.No.	Full Name	Registered Number	Program/Branch/Section
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Summary of Research Project:

The increasing adoption of electric vehicles (EVs) has led to a pressing demand for efficient and sustainable charging infrastructure. This project focuses on developing an Internet of Things (IoT) architecture for battery charging management technology, which integrates renewable energy sources, such as solar power and energy storage systems (ESS), with traditional grid power. The proposed architecture aims to optimize the charging process, reduce operational costs, and enhance user experience through real-time data monitoring and management.

Research Outcome:

1. **Increased Charging Efficiency:** The IoT architecture optimizes the charging process by combining solar, energy storage systems (ESS), and grid power, achieving a 90% reduction in energy costs for electric vehicle (EV) charging stations.
2. **Real-Time Monitoring:** Users can track their vehicle's charging status and manage energy use through a mobile and web application, improving the overall user experience.
3. **Power Demand Prediction:** The system uses analytics to forecast power needs and balance loads effectively, ensuring efficient operation of charging stations and reducing energy waste.
4. **Better Battery Management:** A robust Battery Management System (BMS) monitors battery health and performance, promoting safe charging and extending battery life through optimized charging cycles.

5. **Scalable and Secure System:** The project creates a scalable IoT framework that can grow with increasing EV adoption while incorporating strong security measures to protect data and ensure safe communication between devices.

Future Scope:

1. Optimize operations to significantly lower operational costs by monitoring energy consumption and automating maintenance schedules, preventing costly downtimes.
2. Enables real-time data collection and analysis to streamline processes, improve productivity, and allow for proactive maintenance, minimizing disruptions.
3. Provides customers with real-time information on product availability and personalized recommendations, enhancing engagement and satisfaction, which leads to increased loyalty.
4. Facilitates the collection of valuable data on customer behavior and operational performance, allowing businesses to identify trends and make informed decisions for growth.
5. The scalable IoT architecture allows businesses to expand their charging infrastructure as demand increases, enabling quick adaptation to changing market conditions and consumer needs.

Recommendation by Faculty Member with a signature *(please leave it blank if there is No faculty mentor for this project):*

Recommended the project proposal for ANVESHAN 2024.

